PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 2

CANDIDATE NAME	

CT GROUP	1	7	S				INDEX NUMBER					

CHEMISTRY **9729/01**

Paper 1 Multiple Choice **21 September 2018**

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

For each question, there are four possible answers labelled **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

1 In an experiment, 25.0 cm^3 of 0.02 mol dm^{-3} of vanadium(II) ions was found to react with 15.00 cm^3 of 0.02 mol dm^{-3} of acidified $KMnO_4$. The half equation for the reduction of MnO_4^- is:

$$MnO_4^- + 8H^+ + 5e \rightleftharpoons Mn^{2+} + 4H_2O$$

What is the final oxidation state of vanadium?

A +3

B +4

C +5

D +6

2 Sulfur dichloride dioxide, SO_2Cl_2, reacts with water to give a mixture of sulfuric acid and hydrochloric acid.

How many moles of calcium hydroxide, $Ca(OH)_2$, would be needed to neutralise the solution formed by adding one mole of SO_2Cl_2 to an excess of water?

A 1

B 2

C 3

D 4

3 *The use of Data Booklet is relevant to this question.*

An ion $\mathbf{E}^{2+}$ contains 24 protons.

Which of the following statements about $\mathbf{E}^{2+}$ is **incorrect**?

A The enthalpy change for the reaction $\mathbf{E}(g) \rightarrow \mathbf{E}^{2+}(g) + 2e$ is +2243 kJ mol^{-1}.

B The removal of the two electrons from **E** to form $\mathbf{E}^{2+}$ is from the 4s subshell.

C The angle of deflection of $\mathbf{E}^{2+}$ in an electric field is smaller than that of $\mathbf{E}^{3+}$.

D $\mathbf{E}^{2+}$ is isoelectronic with Mn^{3+}.

4 Which of the following statements explain why aluminium chloride, Al_2Cl_6, sublime at a relatively low temperature?

1 Intermolecular forces between the Al_2Cl_6 molecules are weak.

2 The dative bonds between A*l* and C*l* atoms are weak.

3 The covalent bonds between A*l* and C*l* atoms are weak.

A 1 only

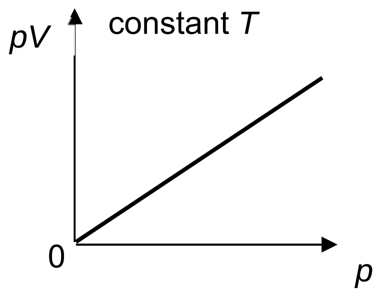
B 1 and 2 only

C 2 and 3 only

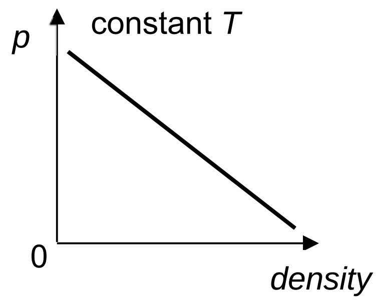
D 3 only

5 Which of the following diagrams correctly describes the behavior of a fixed mass of an ideal gas? (*T* is measured in K)

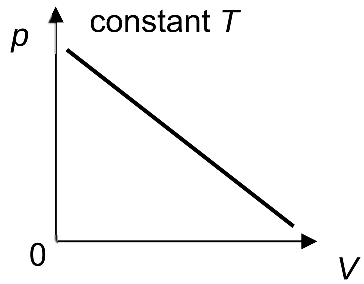
A



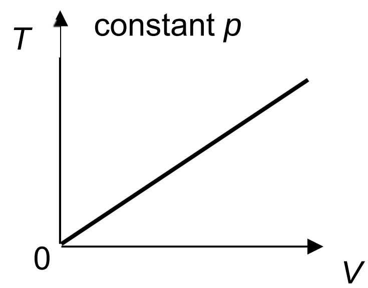
B



C



D



6 The enthalpy change of reaction between hydrogen sulfide and oxygen is ΔH_1.

$$H_2S(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g) + S(s) \qquad \Delta H_1$$

What information is **not** needed to calculate ΔH_1?

A enthalpy change of vaporisation of $H_2O(l)$

B enthalpy change of formation of $H_2S(g)$

C enthalpy change of formation of $H_2O(l)$

D enthalpy change of combustion of S(s)

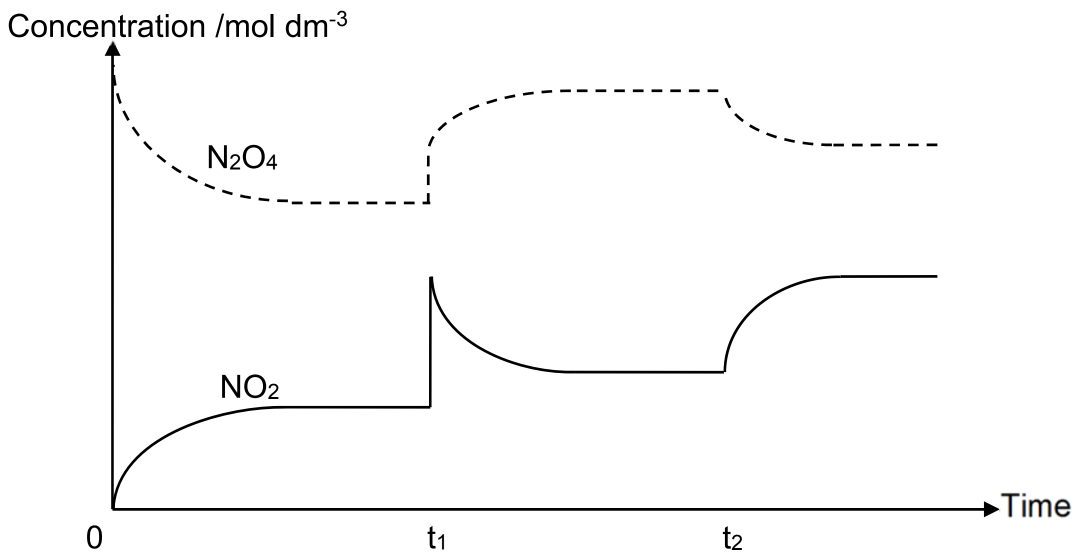
7 Which suggested mechanism is consistent with the experimentally determined rate equations?

	Rate equation	Suggested mechanism	
A	Rate = $k[NO]^2[O_2]$	$2NO(g) \rightarrow N_2O_2(g)$	(fast)
		$N_2O_2(g) + O_2(g) \rightarrow 2NO_2(g)$	(slow)
B	Rate = $k[H_2][I_2]$	$H_2(g) \rightarrow 2H(g)$	(slow)
		$2H(g) + I_2(g) \rightarrow 2HI(g)$	(fast)
C	Rate = $k[HBr][O_2]$	$2HBr(g) + O_2(g) \rightarrow 2HBrO(g)$	(slow)
		$HBrO(g) + HBr(g) \rightarrow H_2O(g) + Br_2(g)$	(fast)
D	Rate = $k[H_2O_2][I^-]$	$2H^+(aq) + 2I^-(aq) \rightarrow 2HI(aq)$	(fast)
		$2HI(aq) + H_2O_2(aq) \rightarrow I_2(aq) + 2H_2O(l)$	(slow)

8 An amount of N_2O_4 was placed in a closed vessel and allowed to reach equilibrium as shown below.

$$N_2O_4(g) \rightleftharpoons 2NO_2(g) \qquad \Delta H = +57 \text{ kJ mol}^{-1}$$

Two changes were made to the equilibrium system at t_1 and t_2.



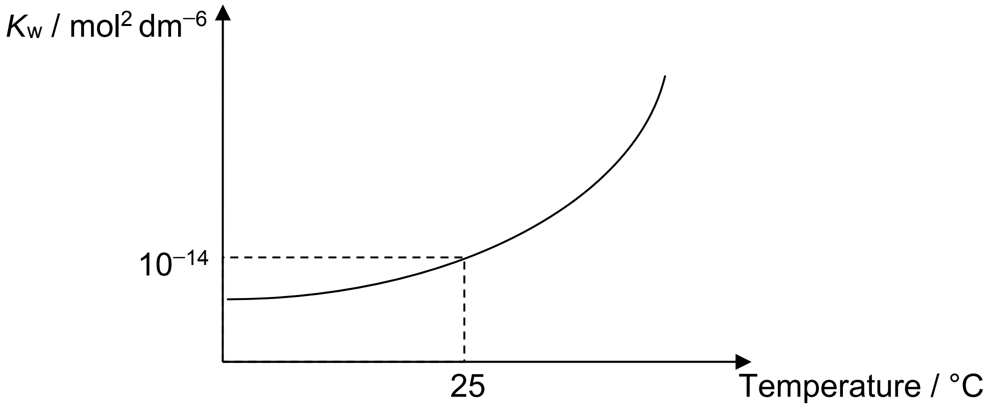
Which are the changes made at t_1 and t_2?

	t_1	t_2
A	More NO_2 was added	Temperature was increased
B	An inert gas was added at constant volume	Temperature was decreased
C	Volume of the system was decreased	Temperature was increased
D	Volume of the system was increased	Temperature was decreased

9 Water dissociates as shown:

$$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$$

The ionic product of water, K_w, varies with temperature as shown in the graph below.



Which statements about the above equilibrium system are correct?

1 The pH of water decreases as temperature increases.

2 The concentrations of H^+ and OH^- are equal at all temperatures.

3 The dissociation of water is exothermic.

A 2 only

B 3 only

C 1 and 2 only

D 2 and 3 only

10 The pH range and colour changes for two acid-base indicators are given below:

Indicator	Colour in acidic solution	pH range	Colour in basic solution
P	Violet	3.0-5.0	red
Q	Yellow	5.7-7.6	blue

Which solution will appear red in **P** and yellow in **Q**?

A 0.1 mol dm^{-3} of HC*l*

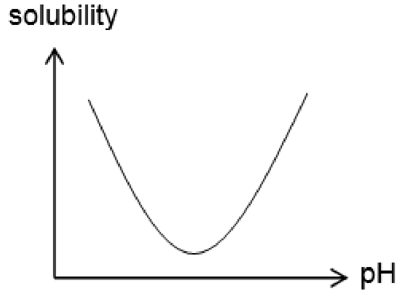
B 0.1 mol dm^{-3} of NaOH

C 0.1 mol dm^{-3} of CH_3CO_2H (K_a = 1.8 x 10^{-5} mol dm^{-3})

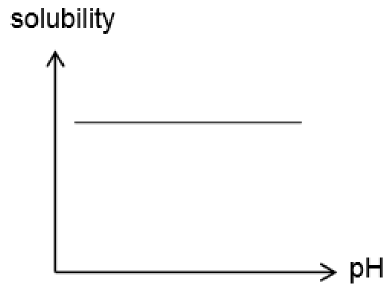
D 0.1 mol dm^{-3} of HX (K_a= 2.5 x 10^{-10} mol dm^{-3})

11 The solubility product of calcium carbonate, $CaCO_3$, is 4.81 x 10^{-9} $mol^2\ dm^{-6}$. Which of the following graphs shows how the solubility of $CaCO_3$ will vary with pH at constant temperature?

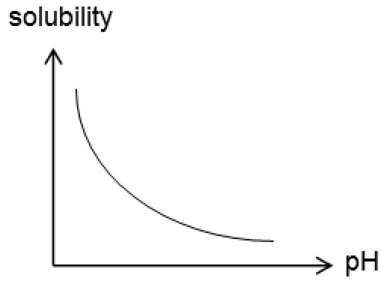
A



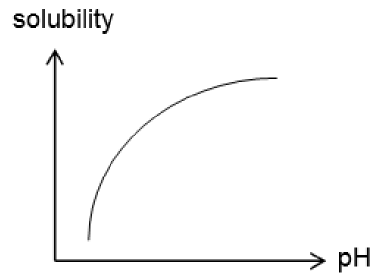
B



C



D



12 The equilibrium between Ag_2CO_3, a sparingly soluble salt, and its saturated solution is as shown below.

$$Ag_2CO_3(s) + aq \rightleftharpoons 2Ag^+(aq) + CO_3^{2-}(aq)$$

K_{sp} of Ag_2CO_3 = 8.2 × 10^{-12} $mol^3\ dm^{-9}$

Which one of the following is correct?

1 Adding Na*Cl*(aq) will cause more Ag_2CO_3 solid to dissolve.

2 Upon the addition of sodium carbonate, solubility of Ag_2CO_3 increases.

3 K_{sp} of Ag_2CO_3 decreases as $AgNO_3$ solution is added to it.

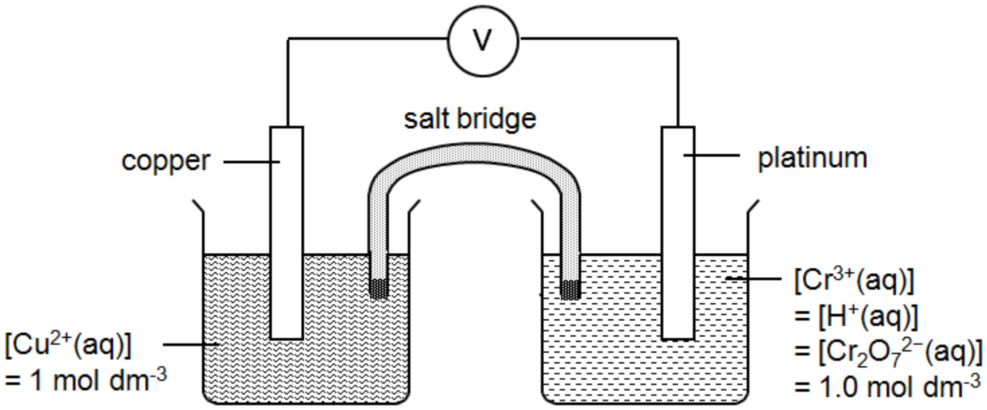
A 1 only

B 1 and 2 only

C 2 and 3 only

D 1, 2 and 3

13 *Use of the Data Booklet is relevant to this question.*
A cell is set up by connecting a Cu^{2+}/Cu half-cell and an acidified $Cr_2O_7^{2-}/Cr^{3+}$ half-cell under standard conditions.

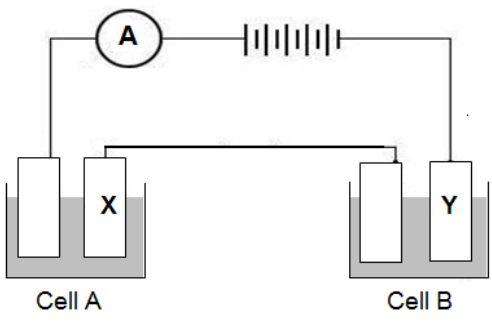


Which of the following correctly describes the effect on the e.m.f of the cell when the respective change is made?

	change	effect on e.m.f of cell
A	using a larger copper electrode	increases
B	addition of concentrated H_2SO_4 into reduction half-cell	decreases
C	addition of dilute NaOH into oxidation half-cell	decreases
D	addition of water into oxidation half-cell	increases

14 *Use of the Data Booklet is relevant to this question.*

A student carried out an experiment involving the electrolysis of aqueous copper(II) sulfate in cell **A** and aqueous chromium(III) sulfate in cell **B**.



Given that 6.35 g of copper was deposited at electrode **X** at the end of the experiment, what is the mass of chromium deposited at electrode **Y**?

A 0.87 g **B** 1.74 g

C 3.47 g **D** 10.4 g

15 ***P*, *Q*** and ***R*** are elements of the third period of the Periodic Table. The oxide of ***P*** is amphoteric, the oxide of ***Q*** is basic and oxide of ***R*** is acidic.

What is the order of increasing ionic radius?

A ***RQP***

B ***QPR***

C ***PRQ***

D ***PQR***

16 Which of the following statements about Group 2 elements and their compounds is correct?

1 Magnesium hydroxide decomposes on heating to give magnesium oxide and steam.

2 1 mole of strontium carbonate, on heating over a short period of time, decomposes to give more carbon dioxide gas than 1 mole of magnesium carbonate

3 Magnesium has a higher melting point than strontium.

A 1 only

B 1 and 3 only

C 2 and 3 only

D 1, 2 and 3

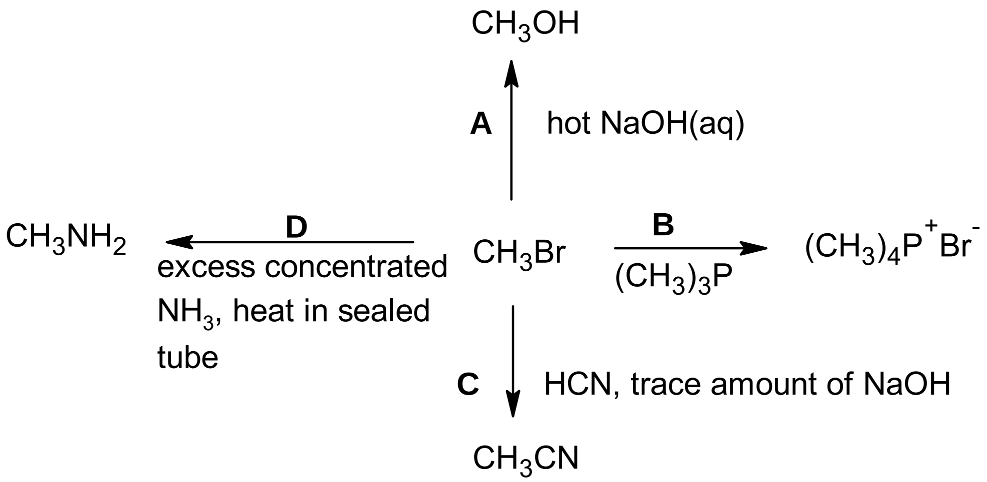
17 Adding KSCN (aq) to $FeNO_3$(aq) causes the colour of the solution to change from yellow to blood red.

Which of the following row correctly shows the number of d-electrons and the energy gap between the d-orbitals, before and after the reaction?

	number of d-electrons	energy gap between the d-orbitals
A	changes	changes
B	changes	remains the same
C	remains the same	changes
D	remains the same	remains the same

18 A reaction scheme involving bromomethane is given below.

Which of the reactions does **not** take place?



19 3-ethylpentane can react with bromine in the presence of sunlight to give two monosubstituted halogenoalkanes:

1-bromo-3-ethylpentane and 3-bromo-3-ethylpentane.

Given the relative rates of abstracting H atoms are:

Type of H atom	primary	secondary	Tertiary
Relative rate of abstraction	1	4	6

What is the expected ratio of 1-bromo-3-ethylpentane to 3-bromo-3-ethylpentane formed?

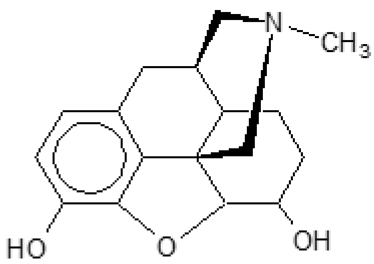
A 9 : 1

B 6 : 1

C 3 : 2

D 1 : 6

20 Morphine is an effective pain killer.



morphine

Which of the following statements about morphine is correct?

A It contains 10 chiral centers.

B It does not decolourise Br_2(aq).

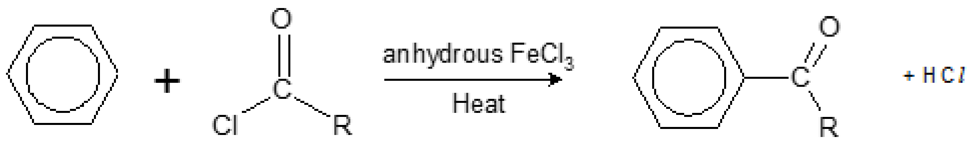
C It turns cold alkaline $KMnO_4$ from purple to colourless.

D One mole of morphine reacts with excess sodium to give 1 mole of H_2 gas.

21 Under suitable conditions, ethene may undergo a reaction with an interhalogen compound, I*Cl*. Which of the following shows the structure of the intermediate formed?

A $[CH_3CH_2]^+$ **B** $[CH_3CHI]^+$ **C** $[CH_2CH_2I]^+$ **D** $[CH_2CH_2Cl]^+$

22 Anhydrous iron(III) chloride is made by passing chlorine gas over heated iron. It can be used as a catalyst in the acylation of benzene, a process called Friedel-Crafts acylation, to produce carbonyl compounds.



The mechanism of the above reaction is similar to that of chlorination of benzene. Which of the following statements is **not** correct about the mechanism?

A The organic intermediate has a ring with one *sp*^{3} hybridised carbon.

B The overall order of reaction is 1.

C The mechanism involves RCO^+ as an electrophile.

D $FeCl_3$ is a homogeneous catalyst.

23 Some chlorobutanes were separately treated with hot ethanolic sodium hydroxide. Two of these gave the same hydrocarbon, C_4H_6.

From which pair of chlorobutanes was this hydrocarbon obtained?

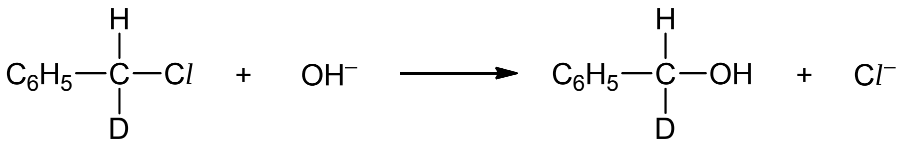
A $CH_3CH_2CH_2CH_2Cl$ and $CH_3CH_2CH_2CHCl_2$

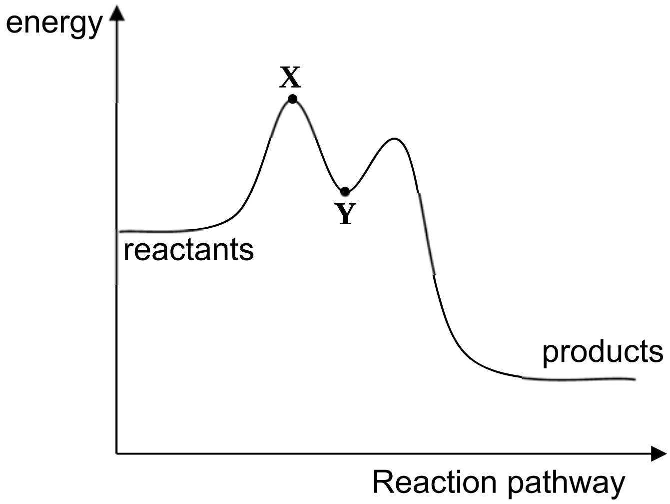
B $CH_3CHClCHClCH_3$ and $ClCH_2CH_2CH_2CH_2Cl$

C $CH_3CH_2CH_2CH_2Cl$ and $ClCH_2CH_2CH_2CH_2Cl$

D $CH_3CH_2CH_2CH_2Cl$ and $CH_3CH_2CHClCH_3$

24 The energy profile for the following reaction is shown below. [D = ^{2}H]



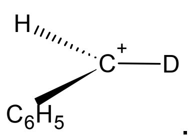


Which of the following conclusions can be drawn?

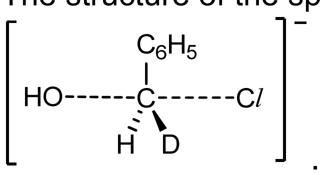
A The product has no effect on the rotation of plane polarised light.

B The rate of reaction can be increased by increasing concentration of OH^-.

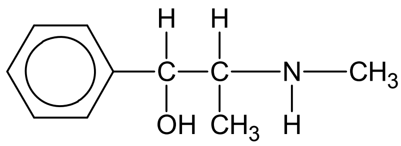
C The structure of the species at point **X** is



D The structure of the species at point **Y** is



25 Ephedrine is a drug that is widely used in cold and allergy medications.

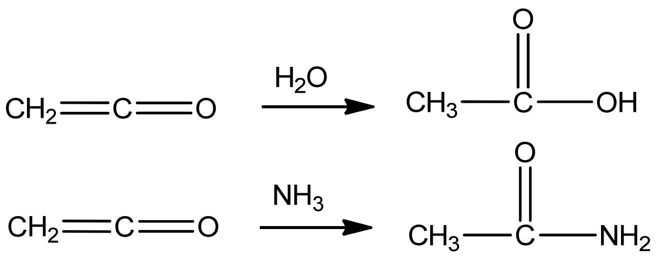


Ephedrine

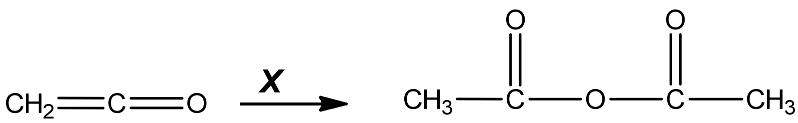
Which of the following pairs of reagents consists of one which reacts with ephedrine and one which does **not** react with ephedrine?

A	CH_3CO_2H	CH_3COCl
B	$HCl(aq)$	$SOCl_2$
C	CH_3Br	$NaOH(aq)$
D	$[Ag(NH_3)_2]^+$	2,4-dinitrophenylhydrazine

26 Ketene ($CH_2{=}C{=}O$) can combine with nucleophiles such as H_2O or NH_3 to make ethanoic acid and ethanamide respectively, as shown below.



Which of the following is the correct nucleophile ***X*** for the reaction below?



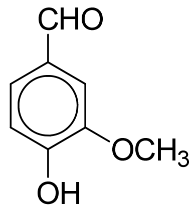
A HCO_2CH_3

B CH_3CHO

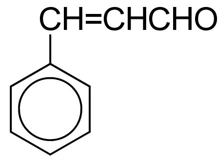
C CH_3COCH_3

D CH_3CO_2H

27 Vanillin and cinnamaldehyde are found in natural products and have very pleasant fragrances.



vanillin



cinnamaldehyde

Which of the following reagents could be used to distinguish between the two compounds? You may assume that the $–OCH_3$ group in vanillin is inert.

1 Fehling's solution

2 hot acidified aqueous $KMnO_4$

3 2, 4-dinitrophenylhydrazine

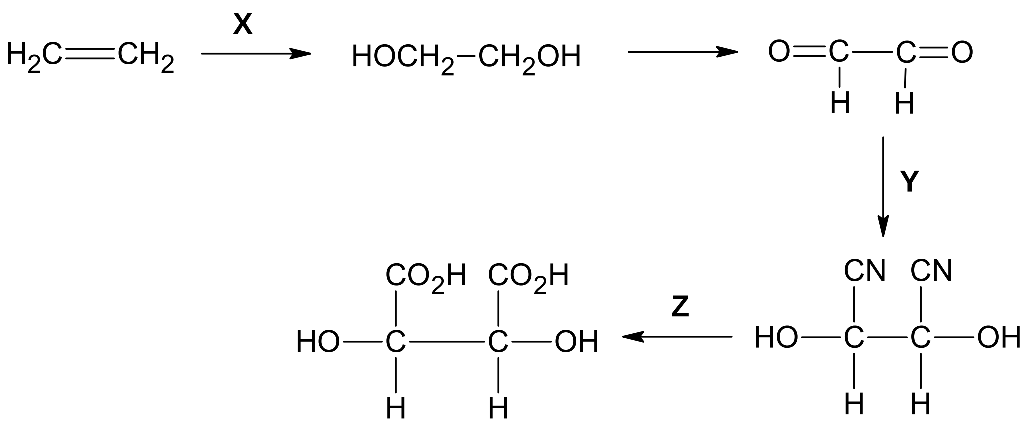
A 1 only

B 2 only

C 1 and 2 only

D 1, 2 and 3

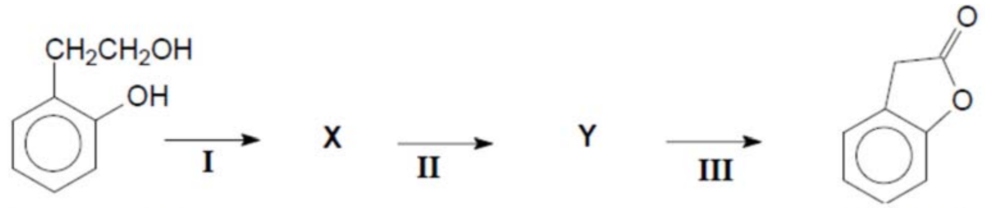
28 The following is a method of synthesising tartaric acid, a compound found in wine.



Which set of reagents and conditions can be used for the synthesis?

	stage **X**	stage **Y**	stage **Z**
A	cold $KMnO_4$, NaOH(aq)	cold HCN, trace KCN(aq)	hot $K_2Cr_2O_7$, H_2SO_4(aq)
B	cold $KMnO_4$, NaOH(aq)	ethanolic KCN, heat	HC*l*(aq), heat
C	cold $KMnO_4$, H_2SO_4(aq)	ethanolic KCN, heat	hot $K_2Cr_2O_7$, H_2SO_4(aq)
D	cold $KMnO_4$, H_2SO_4(aq)	cold HCN, trace KCN(aq)	HC*l*(aq), heat

29 Compound **B** can be converted to compound **C** as shown below.



Compound **B** Compound **C**

Which of the following statements regarding the reaction scheme is correct?

A Step **I** may involve the use of PCl_5.

B Step **I** may involve the use of hot acidified $KMnO_4$.

C Step **III** may involve the use of aqueous NaOH.

D Step **III** may involve the use of hot concentrated H_2SO_4.

30 Which structure will be present when the amino acid aspartic acid,

$$\begin{array}{l} H_2N\,CHCO_2H \\ \quad\quad\;\; | \\ \quad\quad\;\; CH_2CO_2H \end{array}$$

is in aqueous solution at pH 10?

A $\begin{array}{l} H_3\overset{+}{N}-CH-CO_2H \\ \quad\quad\;\;\; | \\ \quad\quad\;\;\; CH_2CO_2H \end{array}$

B $\begin{array}{l} H_3\overset{+}{N}-CH-CO_2^- \\ \quad\quad\;\;\; | \\ \quad\quad\;\;\; CH_2CO_2H \end{array}$

C $\begin{array}{l} H_3\overset{+}{N}-CH-CO_2H \\ \quad\quad\;\;\; | \\ \quad\quad\;\;\; CH_2CO_2^- \end{array}$

D $\begin{array}{l} H_2N-CH-CO_2^- \\ \quad\quad\;\;\; | \\ \quad\quad\;\;\; CH_2CO_2^- \end{array}$

End of paper

2018 P1 Prelim answers

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
C	B	B	A	D	D	A	C	C	D	C	A	D	C	D
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	C	C	C	D	**C**	B	B	A	C	D	C	**D**	C	D